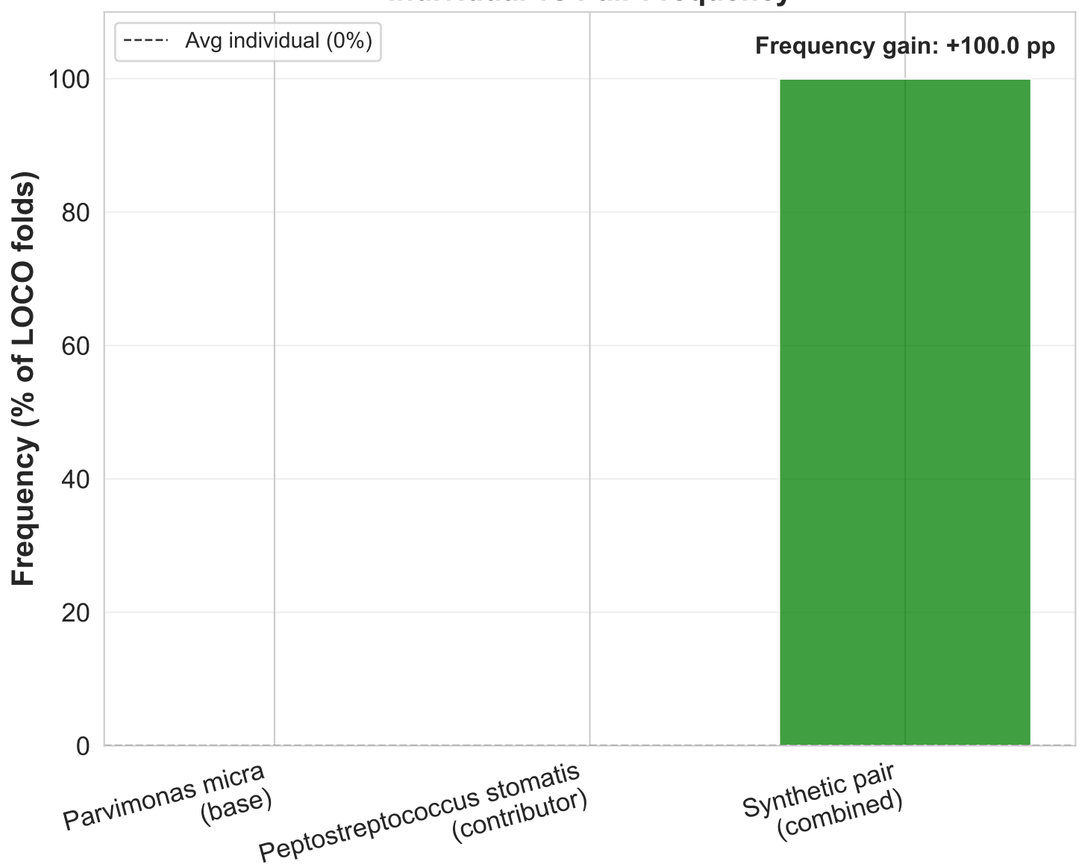
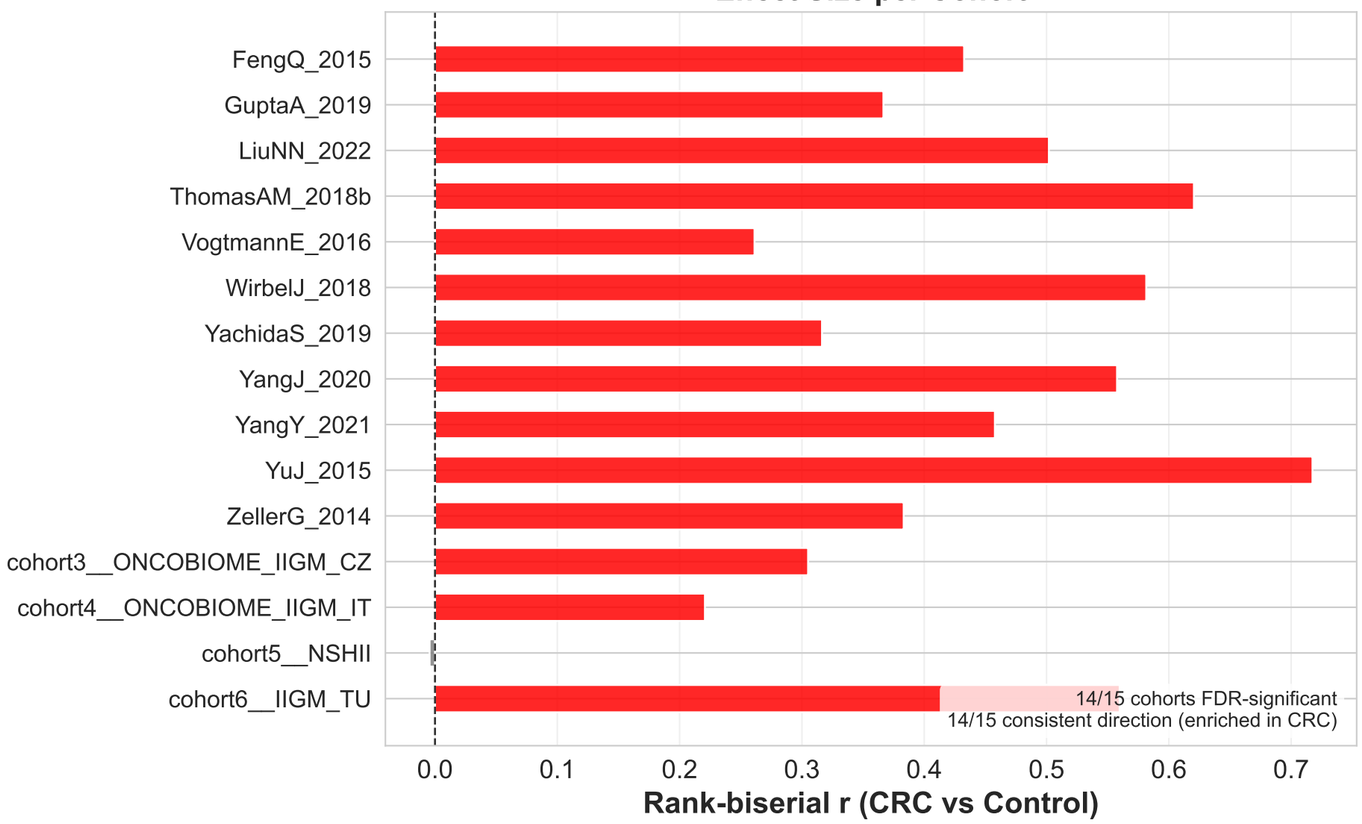
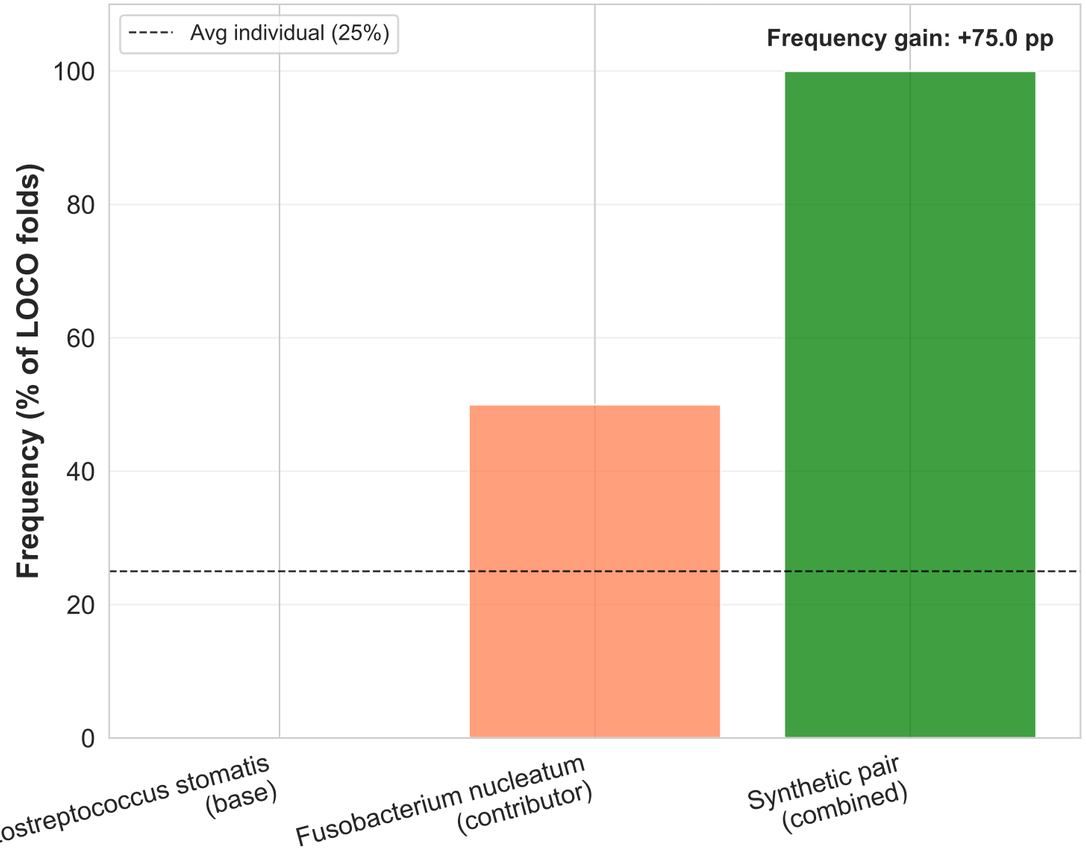
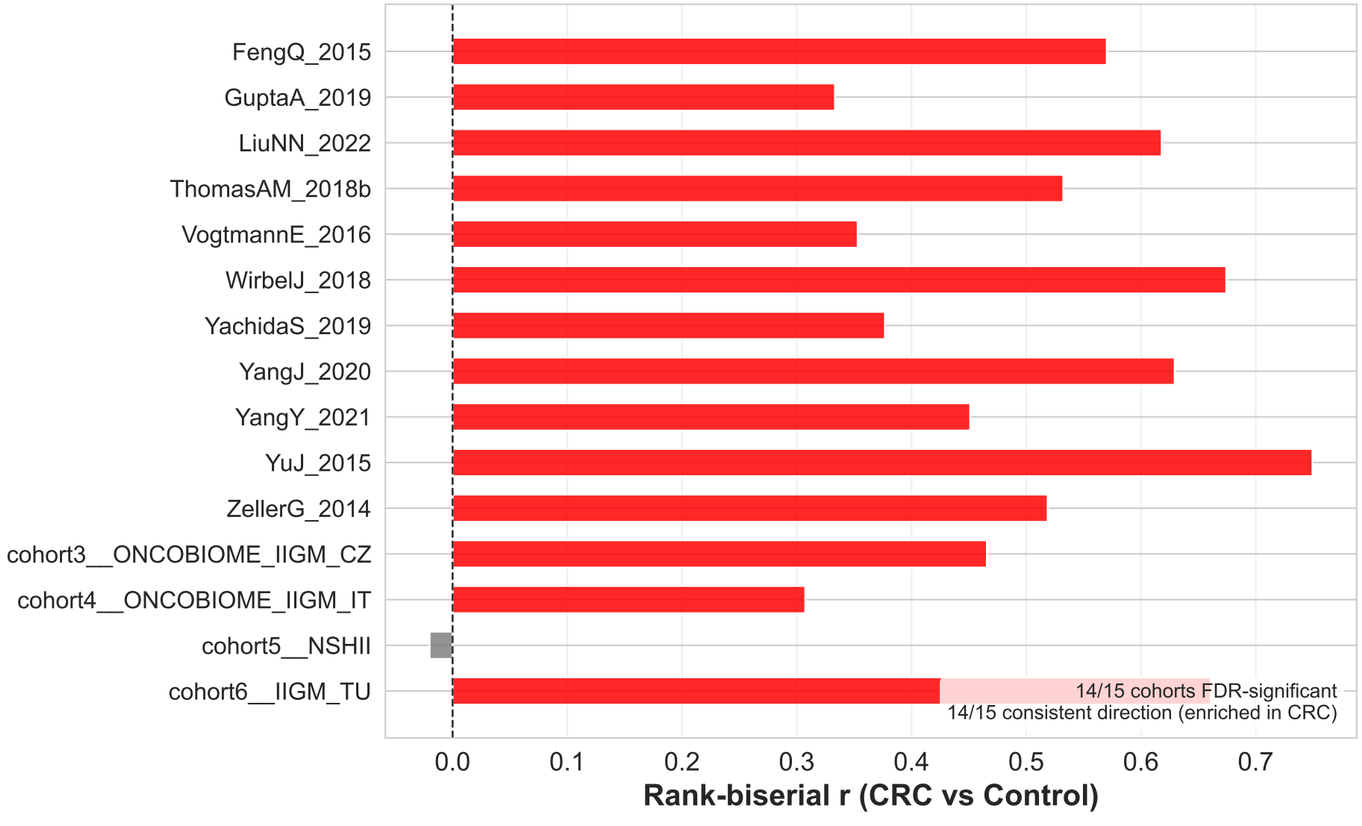
a)


Individual vs Pair Frequency
Avg individual (0%)
Frequency gain: +100.0 pp
Frequency (% of LOCO folds)
0
20
40
60
80
100
Parvimonas micra (base)
Peptostreptococcus stomatis (contributor)
Synthetic pair (combined)


Effect Size per Cohort
FengQ_2015
GuptaA_2019
LiuNN_2022
ThomasAM_2018b
VogtmannE_2016
WirbelJ_2018
YachidaS_2019
YangJ_2020
YangY_2021
YuJ_2015
ZellerG_2014
cohort3__ONCOBIOME_IIGM_CZ
cohort4__ONCOBIOME_IIGM_IT
cohort5__NSHII
cohort6__IIGM_TU
14/15 cohorts FDR-significant
14/15 consistent direction (enriched in CRC)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Rank-biserial r (CRC vs Control)

b)


Individual vs Pair Frequency
Avg individual (25%)
Frequency gain: +75.0 pp
Frequency (% of LOCO folds)
0
20
40
60
80
100
Peptostreptococcus stomatis (base)
Fusobacterium nucleatum (contributor)
Synthetic pair (combined)


Effect Size per Cohort
FengQ_2015
GuptaA_2019
LiuNN_2022
ThomasAM_2018b
VogtmannE_2016
WirbelJ_2018
YachidaS_2019
YangJ_2020
YangY_2021
YuJ_2015
ZellerG_2014
cohort3__ONCOBIOME_IIGM_CZ
cohort4__ONCOBIOME_IIGM_IT
cohort5__NSHII
cohort6__IIGM_TU
14/15 cohorts FDR-significant
14/15 consistent direction (enriched in CRC)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Rank-biserial r (CRC vs Control)
FDR q < 0.05
FDR q >= 0.05